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Energy-optimized cruise control with speed flexibility

Abstract

A cruise control method for a vehicle includes receiving a set speed and determining a commanded axle torque to maintain the set speed on a road that is flat. The method further includes commanding a propulsion system of the vehicle to produce the commanded axle torque to maintain the set speed and applying a fading memory filter to the commanded axle torque to generate a filtered torque command. The method further includes determining an arbitrated torque command as a function of the filtered torque command and command the propulsion system of the vehicle to produce the arbitrated torque command. Also, the method includes determining that a slope of the road is greater than a predetermined slope threshold, setting a temporary set speed, and commanding the propulsion system of the vehicle to maintain the vehicle at the temporary speed.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2012/0309585	12/2011	Whitney	477/3	F16H 63/50
2020/0269689	12/2019	Zebiak et al.	N/A	N/A
2020/0290610	12/2019	Grewal	N/A	B60W 50/0097
2020/0398810	12/2019	Zebiak et al.	N/A	N/A
2021/0245748	12/2020	Zebiak et al.	N/A	N/A
2022/0176827	12/2021	Otanez	N/A	B60K 17/00

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
102020103644	12/2019	DE	N/A
102021111699	12/2021	DE	N/A

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Background/Summary

INTRODUCTION

- (1) The present disclosure relates to a system and method for cruise control.
- (2) This introduction generally presents the context of the disclosure. Work of the presently named inventors, to the extent it is described in this introduction, as well as aspects of the description that may not otherwise qualify as prior art at the time of filing, are neither expressly nor impliedly admitted as prior art against this disclosure.
- (3) Cruise control is currently calibrated to rigidly control a driver's set speed and can be aggressive and inefficient in its attempt to maintain that speed on changes in road grades. This leads to lower fuel economy and unnatural behavior (e.g., aggressive tip-ins and downshifts while going up hills, riding the brakes down hills, etc.).

SUMMARY

- (4) The present disclosure relates to a system and method for cruise control. In an aspect of the present disclosure, the cruise control method for a vehicle includes receiving a set speed, a maximum allowed speed, and a minimum allowed speed. The maximum allowed speed and the minimum allowed speed define an allowed speed range. The method includes determining a commanded axle torque to maintain the set speed on a road that is flat, commanding a propulsion system of the vehicle to produce the commanded axle torque to maintain the set speed on the road

that is flat, monitoring the axle torque command of the conventional PID cruise control system, applying a fading memory filter to this conventional-cruise commanded axle torque to generate a filtered torque command, determining an arbitrated torque command as a function of the filtered torque command, commanding the propulsion system of the vehicle to produce the arbitrated torque command, determining that the present vehicle speed is within a calibratable threshold of the allowed vehicle speed boundary, setting a temporary set speed in response to determining that the present vehicle speed is within a calibratable threshold of the allowed vehicle speed boundary, and commanding the propulsion system of the vehicle to maintain the vehicle at the temporary speed while the vehicle travels along the slope of the road.

(5) In an Aspect of the Present Disclosure, the Arbitrated Torque Command is Based on the Filtered Torque Command Using the Following Equations in Sequential Order:

$\tau_{\text{sub.arb}} = \min[\tau_{\text{sub.maxcomp}}, \tau_{\text{sub.flt}}]$

$\tau_{\text{sub.arb}} = \max[\tau_{\text{sub.arb}}, \tau_{\text{sub.mincomp}}]$ where: $\tau_{\text{sub.arb}}$ is the arbitrated torque command; $\tau_{\text{sub.flt}}$ is the filtered torque command; $\tau_{\text{sub.maxcomp}}$ is a predetermined maximum calibration value; and $\tau_{\text{sub.mincomp}}$ is a predetermined minimum calibration value.

(6) In an aspect of the present disclosure, determining that the slope of the road is greater than a predetermined slope threshold includes detecting an incline on the road that is greater than the slope threshold.

(7) In an aspect of the present disclosure, determining the incline on the road includes determining that an acceleration of the vehicle is less than a predetermined acceleration threshold.

(8) In an aspect of the present disclosure, detecting that the incline on the road is sufficiently high includes determining that a speed of the vehicle is less than a predetermined speed threshold.

(9) In an aspect of the present disclosure, determining that the slope of the road is greater than a predetermined slope threshold includes detecting a decline on the road that is greater than the slope threshold.

(10) In an aspect of the present disclosure, detecting the decline on the road that is greater than the slope threshold includes determining that an acceleration of the vehicle is greater than a predetermined acceleration threshold.

(11) In an aspect of the present disclosure, detecting the decline on the road that is greater than the slope threshold includes determining that a speed of the vehicle is greater than a predetermined speed threshold.

(12) The present disclosure also describes a system including sensors and a controller in communication with the sensors. The controller is programmed to execute the method described above.

(13) The present disclosure also describes a tangible, non-transitory, machine-readable medium, comprising machine-readable instructions, that when executed by a processor, causes the processor to execute the method described above.

(14) Further areas of applicability of the present disclosure will become apparent from the detailed description provided below. It should be understood that the detailed description and specific examples are intended for purposes of illustration only and are not intended to limit the scope of the disclosure.

(15) The above features and advantages, and other features and advantages, of the presently disclosed system and method are readily apparent from the detailed description, including the claims, and exemplary embodiments when taken in connection with the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The present disclosure will become more fully understood from the detailed description and the

accompanying drawings, wherein:

(2) FIG. 1 is a schematic diagram depicting an embodiment of a vehicle including a system for cruise control;

(3) FIG. 2 is a schematic front view of a user interface of the vehicle of FIG. 1;

(4) FIG. 3A is part of a flowchart of a cruise control method;

(5) FIG. 3B is part of the flowchart of the cruise control method of FIG. 3A;

(6) FIG. 3C is part of the flowchart of the cruise control method of FIG. 3A; and

(7) FIG. 3D is part of the flowchart of the cruise control method of FIG. 3A.

DETAILED DESCRIPTION

(8) Reference will now be made in detail to several examples of the disclosure that are illustrated in accompanying drawings. Whenever possible, the same or similar reference numerals are used in the drawings and the descriptions to refer to the same or like parts or steps.

(9) As depicted in FIG. 1, a vehicle **10** generally includes a chassis **12**, a body **14**, front and rear wheels **17** and may be referred to as the host vehicle. The vehicle **10** may be referred to as a motor vehicle. The body **14** is arranged on the chassis **12** and substantially encloses components of the vehicle **10**. The body **14** and the chassis **12** may jointly form a frame. The wheels **17** are each rotationally coupled to the chassis **12** near a respective corner of the body **14**. The vehicle **10** may be an autonomous vehicle, and a cruise control system **89** is incorporated into the vehicle **10**. The cruise control system **89** may alternatively be referred to as a vehicle system. The vehicle **10** is, for example, a vehicle that is automatically controlled to carry passengers from one location to another. The vehicle **10** is depicted in the illustrated embodiment as a passenger car, but it should be appreciated that another vehicle including motorcycles, trucks, sport utility vehicles (SUVs), recreational vehicles (RVs), marine vessels, aircraft, etc., can also be used. The vehicle **10** may be a so-called Level Four or Level Five automation system. A Level Four system indicates “high automation”, referring to the driving mode-specific performance by an automated driving system of the aspects of the dynamic driving task, even if a human driver does not respond appropriately to a request to intervene. A Level Five system indicates “full automation”, referring to the full-time performance by an automated driving system of the aspects of the dynamic driving task under different roadway and environmental conditions that can be managed by a human driver.

(10) The vehicle **10** generally includes a propulsion system **20**, a transmission system **22**, a steering system **24**, a brake system **26**, a sensor system **28**, an actuator system **30**, at least one data storage device **32**, at least one controller **34**, and a communication system **36**. The propulsion system **20** may include an electric machine such as a traction motor and/or a fuel cell propulsion system. The vehicle **10** further includes a battery (or battery pack) **21** electrically connected to the propulsion system **20**. Accordingly, the battery **21** is configured to store electrical energy and to provide electrical energy to the propulsion system **20**. Additionally, the propulsion system **20** may include an internal combustion engine **33** having a plurality of cylinders. When the propulsion system **20** engages active fuel management (AFM), not all of the cylinders of the internal combustion engine **33** are active. Conversely, when the propulsion system disengages AFM, all of the cylinders of the internal combustion engine **33** are active. The transmission system **22** is configured to transmit power from the propulsion system **20** to the vehicle wheels **17** according to selectable speed ratios. The transmission system **22** may include a step-ratio automatic transmission, a continuously-variable transmission, or other appropriate transmission. The brake system **26** is configured to provide braking torque to the vehicle wheels **17**. The brake system **26** may include friction brakes, brake by wire, a regenerative braking system such as an electric machine, and/or other appropriate braking systems. The steering system **24** influences the position of the vehicle wheels **17**. While depicted as including a steering wheel for illustrative purposes, the steering system **24** may not include a steering wheel. The vehicle **10** may include an air-conditioning system **29** with a compressor **31** coupled to the internal combustion engine **33** of the propulsion system **20**. The compressor **31** may be driven by the internal combustion engine **33**.

(11) The sensor system **28** includes one or more sensing devices **40** that sense observable conditions of the exterior environment and/or the interior environment of the vehicle **10**. The sensing devices **40** may include, but are not limited to, radars, lidars, global positioning systems, optical cameras, thermal cameras, ultrasonic sensors, clocks for measuring time, and/or other sensors. The actuator system **30** includes one or more actuator devices **42** that control one or more vehicle features such as, but not limited to, the propulsion system **20**, the transmission system **22**, the steering system **24**, and the brake system **26**. In various embodiments, the vehicle features can further include interior and/or exterior vehicle features such as, but are not limited to, doors, a trunk, and cabin features such as air, music, lighting, etc. (not numbered). The sensing system **28** includes one or more Global Positioning System (GPS) transceiver **40g** configured to detect and monitor the route data (i.e., route information). The GPS transceiver **40g** is configured to communicate with a GPS to locate the position of the vehicle **10** in the globe. The GPS transceiver **40g** is in electronic communication with the controller **34**.

(12) The data storage device **32** stores data for use in automatically controlling the vehicle **10**. In various embodiments, the data storage device **32** stores defined maps of the navigable environment. In various embodiments, the defined maps may be predefined by and obtained from a remote system. For example, the defined maps may be assembled by the remote system and communicated to the vehicle **10** (wirelessly and/or in a wired manner) and stored in the data storage device **32**. As can be appreciated, the data storage device **32** may be part of the controller **34**, separate from the controller **34**, or part of the controller **34** and part of a separate system.

(13) The controller **34** includes at least one processor **44** and a computer non-transitory readable storage device or media **46**. The processor **44** can be a custom made or commercially available processor, a central processing unit (CPU), a graphics processing unit (GPU), an auxiliary processor among several processors associated with the controller **34**, a semiconductor-based microprocessor (in the form of a microchip or chip set), a macroprocessor, a combination thereof, or generally a device for executing instructions. The computer readable storage device or media **46** may include volatile and nonvolatile storage in read-only memory (ROM), random-access memory (RAM), and keep-alive memory (KAM), for example. KAM is a persistent or non-volatile memory that may be used to store various operating variables while the processor **44** is powered down. The computer-readable storage device or media **46** may be implemented using a number of memory devices such as PROMs (programmable read-only memory), EPROMs (electrically PROM), EEPROMs (electrically erasable PROM), flash memory, or another electric, magnetic, optical, or combination memory devices capable of storing data, some of which represent executable instructions, used by the controller **34** in controlling the vehicle **10**. The data storage device **32** and/or the computer readable storage device or media **46** may include a map database **35**. In the present disclosure, the term ‘map database’ means a database that stores geographical and topographical data, such as roads, streets, cities, parks, traffic signs, elevation information, two-dimensional or three-dimensional arrangement of objections with attributes to location and category. The map database **35** includes data about the elevation E of a terrain T_{rr} (FIG. 3A-FIG. 3D) at predetermined-upcoming locations of the vehicle **10**. The data about the elevation E of a terrain T_{rr} (FIG. 3A-FIG. 3D) at the predetermined-upcoming locations of the vehicle **10** is referred herein as upcoming elevation data ED . In the present disclosure, the terrain T_{rr} is the terrain T_{rr} in which the vehicle **10** is traveling or will be traveling. The map database **35** may alternatively be referred to as the map module.

(14) The instructions may include one or more separate programs, each of which comprises an ordered listing of executable instructions for implementing logical functions. The instructions, when executed by the processor **44**, receive and process signals from the sensor system **28**, perform logic, calculations, methods and/or algorithms for automatically controlling the components of the vehicle **10**, and generate control signals to the actuator system **30** to automatically control the components of the vehicle **10** based on the logic, calculations, methods, and/or algorithms.

Although a single controller **34** is shown in FIG. 1, embodiments of the vehicle **10** may include a number of controllers **34** that communicate over a suitable communication medium or a combination of communication mediums and that cooperate to process the sensor signals, perform logic, calculations, methods, and/or algorithms, and generate control signals to automatically control features of the vehicle **10**.

(15) In various embodiments, one or more instructions of the controller **34** are embodied in the cruise control system **89**. The vehicle **10** includes a user interface **23**, which may be a touchscreen in the dashboard. The user interface **23** is in electronic communication with the controller **34** and is configured to receive inputs from a user (e.g., vehicle operator). Accordingly, the controller **34** is configured to receive inputs from the user via the user interface **23**. The user interface **23** includes a display configured to display information to the user (e.g., vehicle operator or passenger).

(16) The communication system **36** is configured to wirelessly communicate information to and from other entities **48**, such as but not limited to, other vehicles (“V2V” communication), infrastructure (“V2I” communication), remote systems, and/or personal devices. In an exemplary embodiment, the communication system **36** is a wireless communication system configured to communicate via a wireless local area network (WLAN) using IEEE 802.11 standards or by using cellular data communication. However, additional or alternate communication methods, such as a dedicated short-range communications (DSRC) channel, are also considered within the scope of the present disclosure. DSRC channels refer to one-way or two-way short-range to medium-range wireless communication channels specifically designed for automotive use and a corresponding set of protocols and standards. Accordingly, the communication system **36** may include one or more antennas and/or transceivers for receiving and/or transmitting signals, such as cooperative sensing messages (CSMs).

(17) FIG. 1 is a schematic block diagram of the cruise control system **89**, which is configured to control the vehicle **10**. The controller **34** of the cruise control system **89** is in electronic communication with the braking system **26**, the propulsion system **20**, and the sensor system **28**. The braking system **26** includes one or more brake actuators (e.g., brake calipers) coupled to one or more wheels **17**. Upon actuation, the brake actuators apply braking pressure on one or more wheels **17** to decelerate the vehicle **10**. The propulsion system **20** includes one or more propulsion actuators for controlling the propulsion of the vehicle **10**. For example, as discussed above, the propulsion system **20** may include internal combustion engine **33** and, in that case, the propulsion actuator may be a throttle specially configured to control the airflow in the internal combustion engine. The sensor system **28** may include one or more accelerometers (or one or more gyroscopes) coupled to one or more wheels **17**. The accelerometer is in electronic communication with the controller **34** and is configured to measure and monitor the longitudinal and lateral accelerations of the vehicle **10**. The sensor system **28** may include one or more speed sensors **40s** configured to measure and monitor the speed (or velocity) of the vehicle **10**. The speed sensor **40s** is coupled to the controller **34** and is in electronic communication with one or more wheels **17**. Accordingly, the controller **34** is programmed to monitor the speed of the vehicle **10** based on the input from the speed sensor **40s**.

(18) FIG. 2 is a schematic diagram of part of the user interface **23**. The vehicle **10** has cruise control, and the driver's set speed **25** (shown in the user interface **23**) can be adjusted by the vehicle occupant with, for example, up/down arrows on the steering wheel of the vehicle **10**. Aside from the driver's set speed **25**, the user interface **23** also shows the speed tolerance **27**, which includes a maximum and minimum speed allowed deviation. The vehicle occupant may adjust the maximum and minimum speed allowed deviation using the user interface **23**. The user interface **23** shows the allowed speed range **37**, which is calculated as a function of the set speed, the maximum allowed speed, and the minimum allowed speed. The maximum allowed speed and the minimum allowed speed are each a speed boundary of an allowed speed range **37**.

(19) With reference to FIGS. 1 and 2, cruise control is currently calibrated to rigidly control to a

driver's set speed, and can be aggressive and inefficient in its attempts to maintain that speed on changing road grades. This leads to lower fuel economy or electric vehicle (EV) efficiency for customers, and unnatural behavior (aggressive tip-ins and downshifts while going up hills, riding the brakes down hills, etc.) The presently disclosed cruise control system **89** delivers smoother, steadier propulsion control and improved energy efficiency by allowing speed to vary within predefined bounds around the driver's set speed, in response to changing road grades. The cruise control system **89** achieves improved results with a novel control strategy that monitors the commands of the existing cruise control algorithm and follows a heavily filtered version of those commands when commanded torque is "close enough" to steady-state road load axle torque, allowing the driver set speed to be maintained if the vehicle encounters small changes in road load (from passenger/cargo loading, wind, or very slight grades). Also, the cruise control system **89** includes logic to detect hills in advance and assign temporary set speeds to prevent holding too high of a speed on inclines or too low of a speed on declines, and logic to better respond to driver inputs on the fly (e.g., tap commands) before allowing energy-optimized control to resume.

(20) FIG. 3A-FIG. 3D illustrate a flowchart of a cruise control method **100**. The method **100** begins at block **102**. At block **102**, the vehicle occupant engages cruise control and inputs the sets speed through the user interface **23**. The vehicle occupant may also adjust the maximum speed and/or minimum speed deviation using the user interface **23**. Because the user interface **23** is in communication with the controller **34**, the controller **34** receives the set speed, and the maximum speed deviation, and the minimum speed deviation. Then, the method **100** continues to block **104**.

(21) At block **104**, the controller **34** determines and stores the minimum allowed speed and the maximum allowed speed. The controller **34** may determine the minimum allowed speed and the maximum allowed speed using the set speed **25**, the minimum speed deviation, and the maximum speed deviation. The maximum allowed speed and the minimum allowed speed define an allowed speed range. Then, the method **100** continues to block **106**. At block **106**, the controller **34** uses a standard cruise control logic to reach the set speed **25**. Further, at block **106**, the controller **34** determines commanded axle torque required to maintain the set speed on a flat road with no wind. Also, the controller **34** commands the propulsion system **20** of the vehicle **10** to produce the commanded axle torque to maintain the set speed on the road that is flat. Then, the controller **34** continues to block **108**.

(22) At block **108**, the controller **34** determines whether each and every of the following conditions have been met, namely: (a) is the current vehicle speed within a predetermined speed threshold (e.g., five miles per hour); (b) has a predetermined amount of time (e.g., 15 seconds) passed since the last tap command; and (c) is the current axle torque within a predetermined torque threshold of the road load nominal axle torque at the set speed. The tap command is a command to increase or decrease the set speed. The vehicle occupant may input a tap command through the user interface **23** or another button. If the not all the conditions are met at block **108**, then the method **100** returns to block **106**. If all the conditions are met at block **108**, then the method **100** continues to block **110**.

(23) At block **110**, the controller **34** enters eco-cruise control. Specifically, the controller **34** monitors the final the final arbitrated torque request (i.e., the commanded axle torque command) from the standard cruise control logic and stores the final arbitrated torque request. Then, the method **100** continues to block **112**. At block **112**, the controller **34** applies a fading memory average filter to the commanded axle torque command with a calibratable factor to generate a filtered torque command. Next, the controller **34** saves the filtered torque command. Then, the method **100** continues to block **114**.

(24) At block **114**, the controller **34** clips the filtered torque command (in case it is sufficiently far away from road load nominal axle torque) using the following equations in a sequential order:

(25) $T_{arb} = \min[T_{maxcomp} , T_{flt}]$ Eq. 1 $T_{arb} = \max[T_{arb} , T_{mincomp}]$ Eq. 2 where:

$\tau_{sub.arb}$ is the arbitrated torque command; $\tau_{sub.flt}$ is the filtered torque command; $\tau_{sub.maxcomp}$ is a predetermined maximum calibration value; and $\tau_{sub.mincomp}$ is a predetermined minimum calibration value.

(26) In other words, at block **114**, the controller **34** determines the arbitrated torque command as a function of the filtered torque command using the equations above. Then, the method **100** continues to block **116**. At block **116**, the controller **34** commands the propulsion system **20** to produce the arbitrated torque command determined at block **114**. Then, the method **100** continues to block **118**.

(27) At block **118**, the controller **34** determines whether the vehicle occupant or a vehicle override has canceled cruise control. If the vehicle occupant or a vehicle override has canceled cruise control, then the method **100** continues to block **120**. At block **120**, the controller **34** uses standard cruise exit and/or disengage control. If the vehicle occupant or a vehicle override has not canceled cruise control, then the method **100** continues to block **122**. At block **122**, the controller **34** determines whether the vehicle occupant has tapped to a different set speed. In other words, the controller **34** determines whether the vehicle occupant has requested a different set speed. If the vehicle occupant has requested a different set speed, then the method **100** returns to block **106**. If the vehicle occupant has not requested a different set speed, then the method **100** continues to block **124**.

(28) At block **124**, the controller **34** determines whether a steep incline has been detected (i.e., determining whether the slope of the road is greater than a predetermined slope threshold). A steep incline is detected if the current vehicle acceleration is less than a predetermined acceleration threshold while the arbitrated torque command is greater than the road load nominal axle torque times a first factor. The first factor is determined through calibration. Alternatively, a steep incline is detected if the current vehicle speed is less than an eco-exit threshold. The eco-exit threshold is equal to a second factor times the minimum allowed speed. The second factor is determined through calibration. If a steep incline is detected, then the method **100** continues to block **142**. If no steep incline has been detected, then the method **100** continues to block **126**.

(29) At block **126**, the controller **34** determines whether a steep decline has been detected (i.e., determining whether the slope of the road is less than a predetermined slope threshold). A steep decline is detected if the current vehicle acceleration is greater than a predetermined acceleration threshold while the arbitrated torque command is less than the road load nominal axle torque times a third factor. The third factor is determined through calibration. Alternatively, a steep decline is detected if the current speed is less than an eco-exit threshold. This eco-exit threshold is equal to a fourth factor times the maximum allowed speed. If no steep decline is detected, then the method **100** proceeds to block **127**. At block **127**, the controller **34** commands the propulsion system **20** to remain in the eco-torque control mode. Further, at block **127**, the controller **34** monitors the final arbitrated torque command from the standard cruise control logic and stores the final arbitrated torque command. After block **127**, the method **100** returns to block **112**. If a steep decline is detected, then the method **100** continues to block **128**.

(30) At block **128**, the controller **34** enters overspeed control. During overspeed control, the controller **34** sets a temporary set speed. The temporary speed is equal to the maximum allowed speed times a fifth factor. The fifth factor is a calibrated value. Further, at block **128**, the controller **34** uses standard, unfiltered cruise control commanded torque to maintain the temporary set speed. Then, the method **100** continues to block **130**.

(31) At Block **130**, the Controller **34** Sets the Arbitrated Cruise Torque Command Using the Following Equations in Sequential Order:

(32) $T_{arb} = \min[0, T_{std}]$ Eq. 3 $T_{arb} = \min[T_{arb}, T_{mincomp}]$ Eq. 4 where: $\tau_{sub.arb}$ is the arbitrated torque command; $\tau_{sub.std}$ is the road load nominal axle torque at the set speed; and $\tau_{sub.mincomp}$ is a predetermined minimum calibration value.

(33) This prevents lingering at the high temporary set speed after the decline has passed. Then, the method **100** proceeds to block **132**. At block **132**, the controller **34** determines whether the vehicle occupant or a vehicle override has canceled cruise control. If the vehicle occupant or a vehicle override has canceled cruise control, then the method **100** continues to block **134**. At block **134**, the controller **34** uses standard cruise exit and/or disengages cruise control. If the vehicle occupant or a vehicle override has not canceled cruise control, then the method **100** continues to block **136**. At block **136**, the controller **34** determines whether the vehicle occupant has tapped to a different set speed. In other words, the controller **34** determines whether the vehicle occupant has requested a different set speed. If the vehicle occupant has requested a different set speed, then the method **100** returns to block **106**. If the vehicle occupant has not requested a different set speed, then the method **100** continues to block **138**.

(34) At block **138**, the controller **34** determines whether each and every of the following conditions have been met, namely: (a) has a predetermined amount of time (e.g., 15 seconds) passed since the steep decline was detected; (b) is the current vehicle speed less than a predetermined speed threshold, wherein the predetermined speed threshold is equal to the minimum allowed speed times a sixth factor; and (c) is the current axle torque greater a predetermined torque threshold, wherein the predetermined torque threshold is equal to a seventh factor times the road load nominal axle torque at the set speed. The sixth factor and the seventh factors are determined through calibration. If all the conditions are met at block **138**, then the method **100** returns to block **110**. If not all the conditions are met at block **138**, then the method **100** continues to block **140**.

(35) At block **140**, the system **89** remains in overspeed control. Further, at block **140**, the controller **34** uses the standard, unfiltered cruise control torque command to maintain the temporary set speed. After block **140**, the method **100** returns to block **130**.

(36) Returning to block **124**, if a steep incline is detected, then the method **100** continues to block **142**. At block **142**, the system **89** enters underspeed control. During underspeed control, the controller **34** sets a temporary set speed. The temporary speed equals the minimum allowed speed times an eight factor. The eight factor may be determined through calibration. Further, the controller **34** uses the standard, unfiltered cruise control torque command to maintain the temporary set speed. Then, the method **100** continues to block **144**.

(37) At block **144**, the controller **34** sets the arbitrated cruise torque command using the following equation:

(38) $T_{arb} = \max[0, T_{std}]$ Eq. 5 where: $\tau_{sub.arb}$ is the arbitrated torque command; and $\tau_{sub.std}$ is the road load nominal axle torque at the set speed.

(39) This prevents lingering at the low temporary set speed after the incline has passed. Then, the method **100** continues to block **146**. At block **146**, the controller **34** determines whether the vehicle occupant or a vehicle override has canceled cruise control. If the vehicle occupant or a vehicle override has canceled cruise control, then the method **100** continues to block **148**. At block **148**, the controller **34** uses standard cruise exit and/or disengage control. If the vehicle occupant or a vehicle override has not canceled cruise control, then the method **100** continues to block **150**. At block **150**, the controller **34** determines whether the vehicle occupant has tapped to a different set speed. In other words, the controller **34** determines whether the vehicle occupant has requested a different set speed. If the vehicle occupant has requested a different set speed, then the method **100** returns to block **106**. If the vehicle occupant has not requested a different set speed, then the method **100** continues to block **152**.

(40) At block **152**, the controller **34** determines whether each and every of the following conditions have been met, namely: (a) has a predetermined amount of time (e.g., 15 seconds) passed since the steep incline was detected; (b) is the current vehicle speed greater than a predetermined speed threshold, wherein the predetermined speed threshold is equal to the minimum allowed speed times a ninth factor; and (c) is the current axle torque less a predetermined torque threshold, wherein the predetermined torque threshold is equal to a tenth factor times the road load nominal axle torque at

the set speed. The ninth factor and the tenth factor are determined through calibration. If all the conditions are met at block **152**, then the method **100** returns to block **110**. If not all the conditions are met at block **152**, then the method **100** continues to block **154**.

(41) At block **154**, the system **89** remains in underspeed control. Further, at block **154**, the controller **34** uses the standard, unfiltered cruise control torque command to maintain the temporary set speed. After block **154**, the method **100** returns to block **144**.

(42) While exemplary embodiments are described above, it is not intended that these embodiments describe all possible forms encompassed by the claims. The words used in the specification are words of description rather than limitation, and it is understood that various changes can be made without departing from the spirit and scope of the disclosure. As previously described, the features of various embodiments can be combined to form further embodiments of the presently disclosed system and method that may not be explicitly described or illustrated. While various embodiments could have been described as providing advantages or being preferred over other embodiments or prior art implementations with respect to one or more desired characteristics, those of ordinary skill in the art recognize that one or more features or characteristics can be compromised to achieve desired overall system attributes, which depend on the specific application and implementation. These attributes can include, but are not limited to cost, strength, durability, life cycle cost, marketability, appearance, packaging, size, serviceability, weight, manufacturability, ease of assembly, etc. As such, embodiments described as less desirable than other embodiments or prior art implementations with respect to one or more characteristics are not outside the scope of the disclosure and can be desirable for particular applications.

(43) The drawings are in simplified form and are not to precise scale. For purposes of convenience and clarity only, directional terms such as top, bottom, left, right, up, over, above, below, beneath, rear, and front, may be used with respect to the drawings. These and similar directional terms are not to be construed to limit the scope of the disclosure in any manner.

(44) Embodiments of the present disclosure are described herein. It is to be understood, however, that the disclosed embodiments are merely examples and other embodiments can take various and alternative forms. The figures are not necessarily to scale; some features could be exaggerated or minimized to display details of particular components. Therefore, specific structural and functional details disclosed herein are not to be interpreted as limiting, but merely as a representative basis for teaching one skilled in the art to variously employ the presently disclosed system and method. As those of ordinary skill in the art will understand, various features illustrated and described with reference to any one of the figures may be combined with features illustrated in one or more other figures to produce embodiments that are not explicitly illustrated or described. The combinations of features illustrated provide representative embodiments for typical applications. Various combinations and modifications of the features consistent with the teachings of this disclosure, however, could be desired for particular applications or implementations.

(45) Embodiments of the present disclosure may be described herein in terms of functional and/or logical block components and various processing steps. It should be appreciated that such block components may be realized by a number of hardware, software, and/or firmware components configured to perform the specified functions. For example, an embodiment of the present disclosure may employ various integrated circuit components, e.g., memory elements, digital signal processing elements, logic elements, look-up tables, or the like, which may carry out a variety of functions under the control of one or more microprocessors or other control devices. In addition, those skilled in the art will appreciate that embodiments of the present disclosure may be practiced in conjunction with a number of systems, and that the systems described herein are merely exemplary embodiments of the present disclosure.

(46) For the sake of brevity, techniques related to signal processing, data fusion, signaling, control, and other functional aspects of the systems (and the individual operating components of the systems) may not be described in detail herein. Furthermore, the connecting lines shown in the

various figures contained herein are intended to represent example functional relationships and/or physical couplings between the various elements. It should be noted that alternative or additional functional relationships or physical connections may be present in an embodiment of the present disclosure.

(47) This description is merely illustrative in nature and is in no way intended to limit the disclosure, its application, or uses. The broad teachings of the disclosure can be implemented in a variety of forms. Therefore, while this disclosure includes particular examples, the true scope of the disclosure should not be so limited since other modifications will become apparent upon a study of the drawings, the specification, and the following claims.

Claims

1. A cruise control method for a vehicle, comprising: receiving a set speed, a maximum allowed speed, and a minimum allowed speed, wherein the maximum allowed speed and the minimum allowed speed define an allowed speed range; determining a commanded axle torque to maintain the set speed on a flat portion of a road; commanding a propulsion system of the vehicle to produce the commanded axle torque to maintain the set speed on the flat portion of the road; applying a fading memory filter to the commanded axle torque to generate a filtered torque command; determining an arbitrated torque command as a function of the filtered torque command; commanding the propulsion system of the vehicle to produce the arbitrated torque command; determining that a slope of a sloped portion of the road is greater than a predetermined slope threshold; setting a temporary set speed in response to determining that the slope of the road is greater than the predetermined slope threshold; and commanding the propulsion system of the vehicle to maintain the vehicle at the temporary speed while the vehicle travels along the sloped portion of the road, and wherein the arbitrated torque command based on the filtered torque command using the following equation in sequential order:

$$\tau_{\text{sub.arb}} = \min[\tau_{\text{sub.maxcomp}}, \tau_{\text{sub.flt}}]$$

$\tau_{\text{sub.arb}} = \max[\tau_{\text{sub.arb}}, \tau_{\text{sub.mincomp}}]$ where: $\tau_{\text{sub.arb}}$ is the arbitrated torque command; $\tau_{\text{sub.flt}}$ is the filtered torque command; $\tau_{\text{sub.maxcomp}}$ is a predetermined maximum calibration value; and $\tau_{\text{sub.mincomp}}$ is a predetermined minimum calibration value.

2. The method of claim 1, wherein determining that the slope of the road is greater than a predetermined slope threshold includes detecting an incline on the road that is greater than the slope threshold.
3. The method of claim 2, wherein detecting the incline on the road includes determining that an acceleration of the vehicle is less than a predetermined acceleration threshold.
4. The method of claim 2, wherein detecting the incline on the road includes determining that a speed of the vehicle is less than a predetermined speed threshold.
5. The method of claim 1, wherein determining that the slope of the road is greater than a predetermined slope threshold includes detecting a decline on the road that is greater than the slope threshold.
6. The method of claim 5, wherein detecting the decline on the road that is greater than the slope threshold includes determining that an acceleration of the vehicle is greater than a predetermined acceleration threshold.
7. The method of claim 5, wherein detecting the decline on the road that is greater than the slope threshold includes determining that a speed of the vehicle is greater than a predetermined speed threshold.
8. A system, comprising: a plurality of sensors; and a controller in communication with the plurality of sensors, wherein the controller is programmed to: receive a set speed, a maximum allowed speed, and a minimum allowed speed, wherein the maximum allowed speed and the minimum allowed speed define an allowed speed range; determining a commanded axle torque to

maintain the set speed on a flat portion of a road; command a propulsion system of a vehicle to produce the commanded axle torque to maintain the set speed on the flat portion of the road; apply a fading memory filter to the commanded axle torque to generate a filtered torque command; determine an arbitrated torque command as a function of the filtered torque command; command the propulsion system of the vehicle to produce the arbitrated torque command; determine that a slope of a sloped portion of the road is greater than a predetermined slope threshold; set a temporary set speed in response to determining that the slope of the sloped portion of the road is greater than the predetermined slope threshold; and command the propulsion system of the vehicle to maintain the vehicle at the temporary speed while the vehicle travels along the sloped portion of the road, wherein the arbitrated torque command based on the filtered torque command using the following equation in sequential order:

$$\tau_{\text{sub.arb}} = \min[\tau_{\text{sub.maxcomp}}, \tau_{\text{sub.flt}}]$$

$\tau_{\text{sub.arb}} = \max[\tau_{\text{sub.arb}}, \tau_{\text{sub.mincomp}}]$ where: $\tau_{\text{sub.arb}}$ is the arbitrated torque command; $\tau_{\text{sub.flt}}$ is the filtered torque command; $\tau_{\text{sub.maxcomp}}$ is a predetermined maximum calibration value; and $\tau_{\text{sub.mincomp}}$ is a predetermined minimum calibration value.

9. The system of claim 8, wherein the controller is programmed to determine that the slope of the road is greater than the predetermined slope threshold by detecting an incline on the road that is greater than the slope threshold.

10. The system of claim 9, wherein the controller is programmed to detect the incline on the road that is greater than the slope threshold by determining that an acceleration of the vehicle is less than a predetermined acceleration threshold.

11. The system of claim 10, wherein the controller is programmed to detect the incline on the road that is greater than the slope threshold by determining that a speed of the vehicle is less than a predetermined speed threshold.

12. The system of claim 8, wherein the controller is programmed to determine that the slope of the road is greater than the predetermined slope threshold by detecting a decline on the road that is greater than the slope threshold.

13. The system of claim 12, wherein the controller is programmed to detect that the decline on the road is greater than the slope threshold includes determining that an acceleration of the vehicle is greater than a predetermined acceleration threshold.

14. The system of claim 12, wherein the controller is programmed to detect that the decline on the road is greater than the slope threshold includes determining that a speed of the vehicle is greater than a predetermined speed threshold.

15. A tangible, non-transitory, machine-readable medium, comprising machine-readable instructions, that when executed by a processor, cause the processor to: receive a set speed, a maximum allowed speed, and a minimum allowed speed, wherein the maximum allowed speed and the minimum allowed speed define an allowed speed range; determine a commanded axle torque to maintain the set speed on a flat portion of a road; command a propulsion system of a vehicle to produce the commanded axle torque to maintain the set speed on the flat portion of the road; apply a fading memory filter to the commanded axle torque to generate a filtered torque command; determine an arbitrated torque command as a function of the filtered torque command; command the propulsion system of the vehicle to produce the arbitrated torque command; determine that a slope of a sloped portion of the road is greater than a predetermined slope threshold; set a temporary set speed in response to determining that the slope of the sloped portion of the road is greater than the predetermined slope threshold; and command the propulsion system of the vehicle to maintain the vehicle at the temporary speed while the vehicle travels along the sloped portion of the road, wherein the arbitrated torque command based on the filtered torque command using the following equation in sequential order:

$$\tau_{\text{sub.arb}} = \min[\tau_{\text{sub.maxcomp}}, \tau_{\text{sub.flt}}]$$

$\tau_{\text{sub.arb}} = \max[\tau_{\text{sub.arb}}, \tau_{\text{sub.mincomp}}]$ where: $\tau_{\text{sub.arb}}$ is the arbitrated torque command;

$\tau_{sub.fit}$ is the filtered torque command; $\tau_{sub.maxcomp}$ is a predetermined maximum calibration value; and $\tau_{sub.mincomp}$ is a predetermined minimum calibration value.

16. The tangible, non-transitory, machine-readable medium of claim 15, wherein the controller is programmed to determine that the slope of the road is greater than the predetermined slope threshold by detecting an incline on the road that is greater than the slope threshold.

17. The tangible, non-transitory, machine-readable medium of claim 15, wherein the controller is programmed to detect the incline on the road that is greater than the slope threshold by determining that an acceleration of the vehicle is less than a predetermined acceleration threshold.
